Quantitative Risk Assessment GIS Tool User Manual

The Quantitative Risk Assessment GIS Tool was developed as part of the NCHRP 23-24 project, Methods to Allow Agencies to Incorporate Quantitative Risk Assessment at Project and Network Level. This tool enables users to view risks associated with various types of transportation assets under different natural threats. See below for instructions on how to use the tool. Please note that the tool is intended for research and planning purposes only.

The input for this GIS tool comes from risk assessment results generated by the QRA Spreadsheet Tool. These results were visualized using ArcGIS Pro and shared via ArcGIS Online. To make the risk information more accessible, the results were also presented through an ArcGIS StoryMap, which constitutes the Quantitative Risk Assessment GIS Tool. The current version of the tool is based on a case study in Utah.

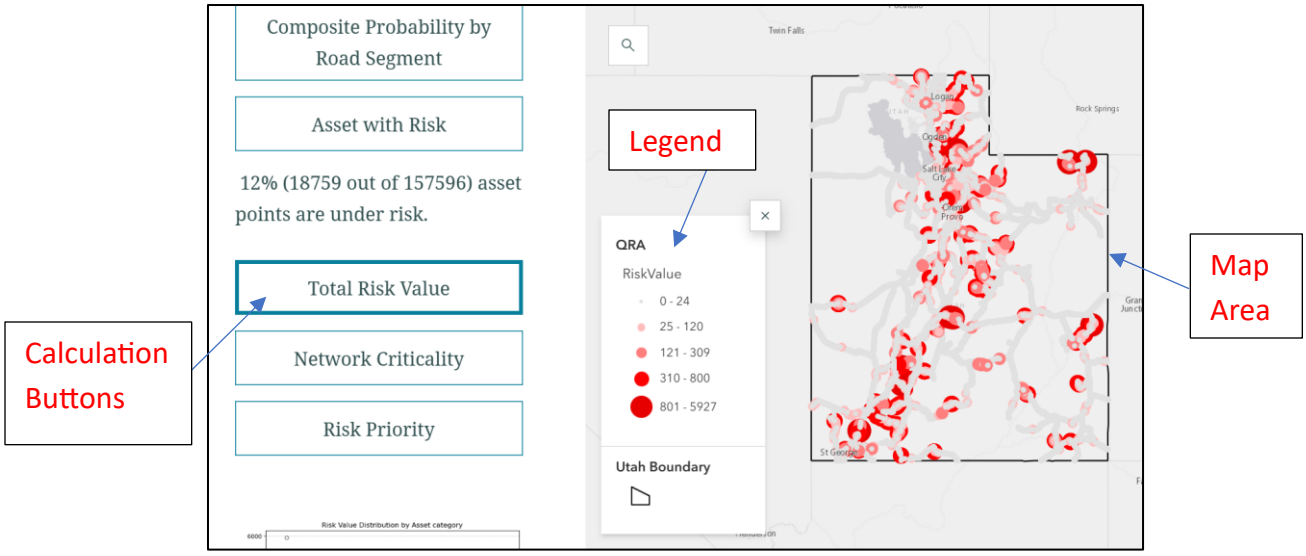
- **Instruction: Click the buttons to view the maps**
  This tool includes multiple maps that display the risk assessment results. To view the map or visualization for a specific calculation, click the corresponding button located on the left side of the map area.

- **Asset Category**
  This section lists the primary transportation assets included in the risk assessment.

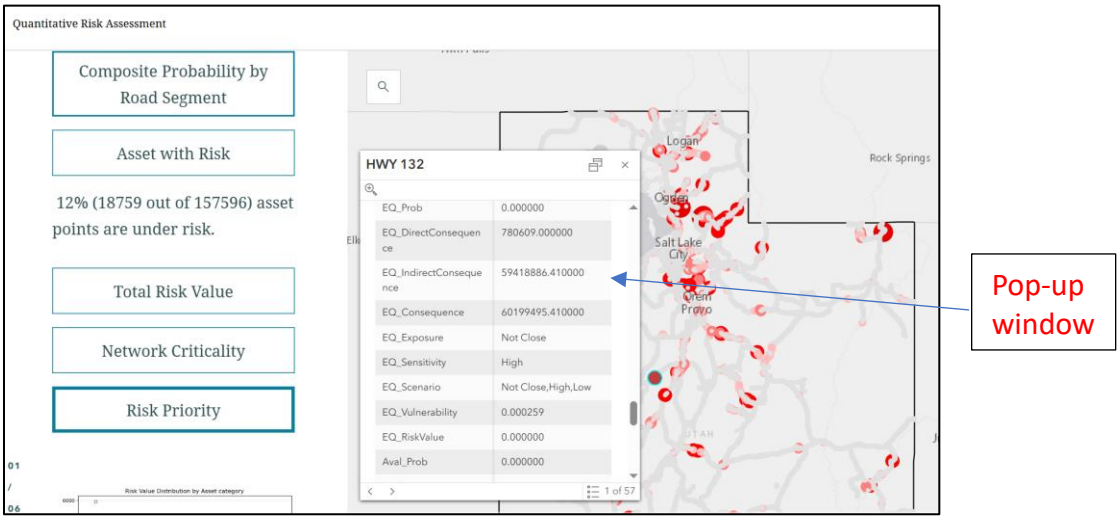
- **Threat Category**
  This section lists the main natural threats considered in the risk assessment.

- **View Asset Risk Information**

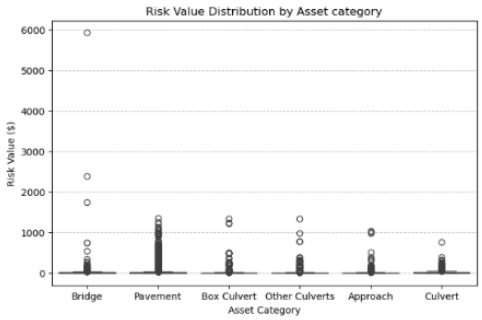
  - **Map Area:** Maps are displayed in the map area on the right side of the webpage. When you click a button, the corresponding map will appear. Each map includes a legend for reference. You can zoom in and out to explore the map in more detail. Right-clicking on a feature will activate a pop-up window with additional information about the selected asset.



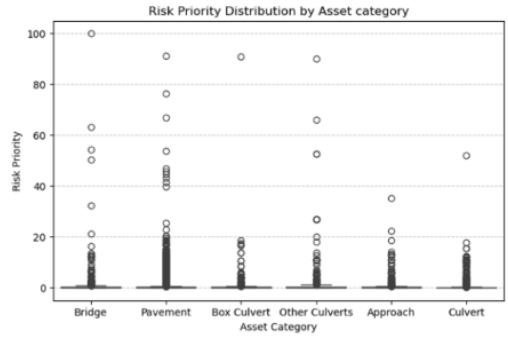
- **Pop-Up Windows:** Right-click on a specific asset on the map to view a pop-up window displaying the asset's attributes and risk assessment results. These include the risk value, risk priority, network criticality, vulnerability, probability of threat, and consequence.



- **Summary Statistics:** This tool also includes several figures and tables that present descriptive statistics for risk values and risk priorities. These visualizations help users understand overall trends and the distribution of risk across different assets and threat categories.



Risk Value by Asset Category



Risk Priority by Asset Category

Asset Category	count	mean	std	min	25%	50%	75%	max
Pavement	11630	43.8419	119.221	0.00327243	1.60182	4.7674	12.8235	1351.45
Other Culverts	652	41.869	108.396	0.00371494	0.93031	2.94939	10.4419	1335.09
Culvert	4798	14.3752	29.1069	0.000355638	0.341755	2.09203	20.0112	772.782
Bridge	604	35.263	274.667	0.0632758	0.608247	1.6237	15.8959	5926.72
Box Culvert	367	49.1436	151.217	0.00919129	2.43373	5.20313	11.4099	1348.18
Approach	708	17.3788	76.4265	0.0282675	0.30626	1.63818	9.83964	1035.64

Summary Statistics of Risk Value

Asset Category	count	mean	std	min	25%	50%	75%	max
Approach	708	0.620986	2.35425	0	0.00564169	0.0501312	0.225322	35.3087
Box Culvert	367	1.01767	5.29976	2.42797e-05	0.0155055	0.0742986	0.27315	91.0477
Bridge	604	1.16545	6.11719	0	0.00929854	0.0609225	0.353704	100
Culvert	4798	0.269654	1.23242	0	0.00231705	0.022553	0.183919	52.189
Other Culverts	652	1.48107	6.12735	0	0.00740681	0.051813	0.486228	90.1636
Pavement	11630	0.580017	2.3602	0	0.00919712	0.0477063	0.259225	91.2689

Summary Statistics of Risk Priority

❖ **Risk Assessment Groups**

This tool includes both the total risk values and risk priorities aggregated across all threat types, as well as detailed risk assessment results for each individual threat. The results for each threat type include the consequence, probability, vulnerability under that specific threat, and the resulting risk value.

Total Risk Value

Network Criticality

Risk Priority

Consequence of Earthquake

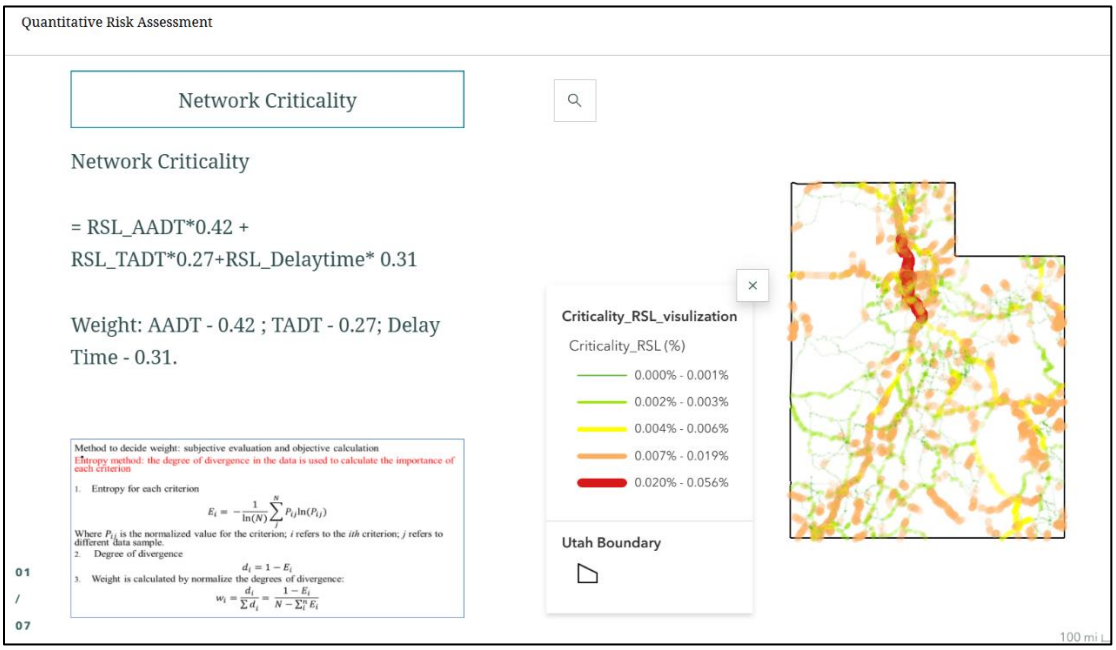
Probability of Earthquake

Vulnerability under Earthquake

Risk Value due to Earthquake

❖ **Network Criticality**

This tool also provides detailed information on network criticality, including the formula and calculation steps used. The map visualizes network criticality and highlights key contributing factors such as AADT (Annual Average Daily Traffic), TADT (Truck Annual Daily Traffic), and delay time.



Contact Information:
Junmei Cheng | UCLA Mobility Lab | junmeicheng@ucla.edu